Mean-Field Limits and Propagation of Chaos

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1 Quantitative Mean-Field Limits via Dobrushin's Estimate

1.1 Overview and Examples

1.1.1 Microscopic, Mesoscopic, and Macroscopic Descriptions

Remark 1.1 (Microscopic, mesoscopic, and macroscopic descriptions). A particle system describes the microscopic state:

$$(X_1(t), \dots, X_N(t)).$$

A kinetic or mean-field equation describes the mesoscopic distribution of a typical particle:

$$f(t,x,v) \qquad \text{or} \qquad \rho_t(x).$$

Macroscopic equations describe coarse observables such as density, momentum, or velocity fields.

A side note: the word "density" can appear at both the mesoscopic and macroscopic levels. At the mesoscopic level, $f(t,x,v)$ is the full one-particle phase-space density. A macroscopic density is usually a coarse observable obtained by taking moments, for instance

$$\varrho(t,x) = \int_{\mathbb{R}^d} f(t,x,v)\,dv.$$

Thus the distinction is not the word "density" itself, but the amount of microscopic information retained. Mean-field limits are one rigorous mechanism for passing from the microscopic particle description to a mesoscopic continuum description.

The goal of a mean-field limit is to reduce a high-dimensional particle description to a lower-dimensional continuum description by looking at averaged quantities:

$$\text{particle model} \qquad \longrightarrow \qquad \text{kinetic or continuum model.}$$

1.1.2 Motivational Examples

Mean-field and kinetic descriptions appear in many different contexts. We list representative examples here; some of them will be revisited in Section 1.1.4 from the structural mean-field viewpoint.

- **Gas dynamics.** Gas molecules interacting by collisions lead to the Boltzmann equation:

  $$\partial_t f + v\cdot\nabla_x f = Q(f,f).$$

  Here $f = f(t,x,v)$ is the density of particles at position x with velocity v. The collision operator $Q(f,f)$ acts only in the velocity variable and has the gain–loss form

  $$Q(f,f)(t,x,v) = \int_{\mathbb{R}^d}\int_{\mathbb{S}^{d-1}} B(|v-v_*|,\cos\theta)\big[f(t,x,v')f(t,x,v'_*) - f(t,x,v)f(t,x,v_*)\big]\,d\sigma\,dv_*,$$

  where v_* is the velocity of the collision partner, $\sigma\in\mathbb{S}^{d-1}$ is the scattering direction, and

  $$v' = \frac{v+v_*}{2} + \frac{|v-v_*|}{2}\sigma, \qquad v'_* = \frac{v+v_*}{2} - \frac{|v-v_*|}{2}\sigma, \qquad \cos\theta = \frac{v-v_*}{|v-v_*|}\cdot\sigma.$$

  The function $B(|v-v_*|,\cos\theta)$ is the collision kernel (cross-section). It determines the rate at which particles with relative velocity $v-v_*$ scatter in direction σ. The term $f(t,x,v')f(t,x,v'_*)$ is the gain term and $f(t,x,v)f(t,x,v_*)$ is the loss term.

- **Plasma physics.** Ions and electrons interacting through Coulomb forces lead to Vlasov–Poisson-type equations:

  $$\partial_t f + v\cdot\nabla_x f + E\cdot\nabla_v f = 0, \qquad E = -\nabla_x\phi, \qquad -\Delta\phi = \rho_f - \rho_{\mathrm{bg}}.$$

- **Atomic physics.** Mean-field limits of many-body quantum systems lead to Hartree-type equations:

  $$i\partial_t\psi = -\Delta\psi + (V*|\psi|^2)\psi.$$

- **Fluid mechanics.** Point vortices in two-dimensional incompressible fluids lead to the vorticity formulation of the Euler equation:

$$\partial_t \omega + \nabla \cdot (u\omega) = 0, \qquad u = K * \omega.$$

- **Opinion dynamics.** Opinions of individuals in a social network can be modeled by interacting particles in an opinion space:

$$\dot{x}_i = \frac{1}{N}\sum_{j=1}^{N} a(|x_i - x_j|)(x_j - x_i).$$

- **Chemotaxis.** Bacteria communicating through chemical signals lead to chemotaxis-type models, for example the Keller–Segel system:

$$\partial_t \rho = \Delta\rho - \nabla \cdot (\rho \nabla c), \qquad -\Delta c + c = \rho.$$

- **Socio-economic models.** Agents participating in a market economy can lead to kinetic models for wealth distributions:

$$\partial_t f = Q_{\mathrm{trade}}(f, f),$$

  where Q_{trade} encodes binary exchanges of wealth.

- **Swarming and collective behavior.** Animals or people interacting through attraction, repulsion, and alignment lead to swarming and crowd dynamics:

$$\dot{x}_i = v_i, \qquad \dot{v}_i = \frac{1}{N}\sum_{j=1}^{N} H(|x_i - x_j|)(v_j - v_i) - \frac{1}{N}\sum_{j=1}^{N} \nabla W(x_i - x_j).$$

- **Optimization and sampling.** Particle optimization and sampling algorithms can often be interpreted as interacting particle systems. A typical consensus-based optimization model has the form

$$dX_i = -\lambda\big(X_i - m_\alpha(\mu_t^N)\big)\,dt + \sigma|X_i - m_\alpha(\mu_t^N)|\,dW_i,$$

  where $m_\alpha(\mu_t^N)$ is a weighted average of the particles.

1.1.3 The Basic Mean-Field Form

Let $z_i(t) \in \mathbb{R}^d$ denote the state of particle i. Depending on the model, z_i may represent a position x_i, a phase-space variable (x_i, v_i), an opinion, a wealth variable, or another state variable.

A basic interacting particle system has the form

$$\dot{z}_i^N(t) = \frac{1}{N}\sum_{j=1}^{N} K\big(z_i^N(t), z_j^N(t)\big), \qquad i = 1, \dots, N. \tag{1}$$

Empirical measure. For any particle configuration $z^N(t) = (z_1^N(t), \dots, z_N^N(t)) \in (\mathbb{R}^d)^N$, define the *empirical measure*

$$\mu_t^N := \frac{1}{N}\sum_{i=1}^N \delta_{z_i^N(t)} \in \mathcal{P}(\mathbb{R}^d).$$

More generally, given a static configuration $X = (x_1, \dots, x_N)$, we write

$$\mu_X^N := \frac{1}{N}\sum_{i=1}^N \delta_{x_i}.$$

This is a probability measure, but it is singular with respect to Lebesgue measure: it consists of N Dirac masses.

The interaction sum can be rewritten as a convolution against μ_t^N:

$$\dot{z}_i^N(t) = (K * \mu_t^N)(z_i^N(t)), \qquad (K * \mu)(z) := \int_{\mathbb{R}^d} K(z, z')\, \mu(dz').$$

Remark 1.2 (Why the empirical measure is the right object)*.* The empirical measure does more than simply record particle locations. It turns the N-particle ODE system into a single evolution equation on the space of probability measures. Although μ_t^N is a sum of Dirac masses and hence is not a classical density, it can still satisfy a PDE in the weak sense, as we will see in Section 1.2.

The mean-field question. The mean-field limit asks:

> When N is large, can we replace the full N-body system by a self-consistent one-particle equation?

The expected limiting equation is the nonlinear continuity equation

$$\partial_t \mu_t + \nabla_z \cdot \big((K * \mu_t)\mu_t\big) = 0. \tag{2}$$

The velocity field is generated by the evolving measure itself.

Key message. The mean-field approximation replaces the N-body problem by a self-consistent one-particle problem. The price is nonlinearity in the law.

Initial sampling and the mean-field limit. Let $\rho_0 \in \mathcal{P}(\mathbb{R}^d)$, and let ρ_t be the corresponding solution of (2). Take N i.i.d. samples

$$z_1^0, \dots, z_N^0 \sim \rho_0,$$

and form the initial empirical measure

$$\mu_0^N = \frac{1}{N}\sum_{i=1}^N \delta_{z_i^0}.$$

By the law of large numbers, $\mu_0^N \rightharpoonup \rho_0$ as $N \to \infty$. Letting $z^N(t)$ be the unique solution of (1) with initial data $(z_1^0, \dots, z_N^0)$, we say that (2) is the *mean-field limit* of (1) if

$$\mu_t^N \rightharpoonup \rho_t \qquad \text{as } N \to \infty,$$

in an appropriate topology.

Remark 1.3. The exact notion of convergence depends on the model. For smooth interaction kernels, one often proves convergence in Wasserstein distance. For rough or singular kernels, one may need weaker notions of convergence, compactness methods, entropy methods, or modulated-energy estimates.

Stochastic version. A stochastic analogue is the additive-noise interacting particle system

$$dz_i^N(t) = \frac{1}{N}\sum_{j=1}^{N} K\big(z_i^N(t), z_j^N(t)\big)\,dt + \sqrt{2\sigma}\,dW_i(t), \qquad i = 1, \dots, N.$$

The corresponding mean-field equation is the nonlinear Fokker–Planck equation

$$\partial_t \mu_t + \nabla_z \cdot \big((K * \mu_t)\mu_t\big) = \sigma \Delta_z \mu_t.$$

Equivalently, the limiting nonlinear process satisfies

$$d\bar{z}(t) = (K * \mu_t)(\bar{z}(t))\,dt + \sqrt{2\sigma}\,dW(t), \qquad \mu_t = \mathrm{Law}(\bar{z}(t)).$$

In this lecture we focus on the deterministic case $\sigma = 0$, where the main tool is Dobrushin's stability estimate. The stochastic version reappears later in connection with propagation of chaos, synchronous coupling, and entropy methods.

Remark 1.4. The additive-noise model above is the simplest stochastic example. In kinetic models, the noise may act only on the velocity variable; in other models, the diffusion may be degenerate, anisotropic, or state-dependent. The main mean-field idea is the same, but the limiting PDE and proof methods must be adjusted accordingly.

1.1.4 Examples of the General Mean-Field Form

We now revisit several models from the structural perspective of (1)–(2).

Autonomous transport. Let $z_i = x_i$ and take $K(z, z') = b(z)$. Then $\dot{x}_i(t) = b(x_i(t))$, so there is no coupling between particles. The mean-field equation becomes the linear continuity equation

$$\partial_t \mu_t + \nabla \cdot (b\mu_t) = 0.$$

Newtonian dynamics and the Vlasov equation. Let $z = (x, v) \in \mathbb{R}^d \times \mathbb{R}^d$. For a single particle, Newtonian dynamics can be written as

$$\dot{x}(t) = v(t), \qquad \dot{v}(t) = \frac{1}{m} F(x(t)).$$

Equivalently, $\dot{z}(t) = b(z(t))$ with $b(x, v) = (v, m^{-1}F(x))$. The corresponding transport equation in phase space is

$$\partial_t f + v \cdot \nabla_x f + \tfrac{1}{m} F(x) \cdot \nabla_v f = 0.$$

This becomes a Vlasov-type equation when the force field is self-consistent.

Discrete aggregation equation. Let $z_i = x_i$ and take $K(z, z') = -\nabla W(z - z')$. The particle system becomes

$$\dot{x}_i(t) = -\frac{1}{N}\sum_{j=1}^{N} \nabla W(x_i(t) - x_j(t)),$$

and the corresponding continuum equation is

$$\partial_t \rho_t = \nabla \cdot \big(\rho_t \, \nabla W * \rho_t\big).$$

Two-dimensional incompressible Euler equation. The incompressible Euler equation in two dimensions is

$$\partial_t u + (u \cdot \nabla)u + \nabla p = 0, \qquad \nabla \cdot u = 0.$$

Taking the curl gives the vorticity formulation

$$\partial_t \omega + \nabla \cdot (u\omega) = 0, \qquad \omega = \partial_{x_1} u_2 - \partial_{x_2} u_1.$$

The velocity is recovered from the vorticity by the Biot–Savart law $u = K * \omega$, where K is the orthogonal gradient of the fundamental solution of the Laplacian in two dimensions. Hence

$$\partial_t \omega + \nabla \cdot ((K * \omega)\omega) = 0,$$

which has the same structural form as the aggregation or mean-field continuity equation.

Remark 1.5 (Point vortices). Onsager's idea of statistical hydrodynamics views two-dimensional turbulence through point vortices. In this interpretation, the vorticity equation can be seen as a mean-field limit of a system of point vortices. This idea is also the basis of vortex methods for numerical approximations of the two-dimensional Euler equation.

Second-order swarming models. In first-order models, particles are described only by their positions $x_i(t)$. In second-order swarming models, such as the Cucker–Smale model, particles carry both position and velocity: $(x_i(t), v_i(t)) \in \mathbb{R}^d \times \mathbb{R}^d$. A typical alignment model has the form

$$\dot{x}_i = v_i, \qquad \dot{v}_i = \frac{1}{N} \sum_{j=1}^{N} H(|x_i - x_j|)(v_j - v_i).$$

The corresponding mean-field limit is a kinetic equation in phase space (x, v), not merely a continuity equation in x.

Remark 1.6. We will not derive this model in class, but it is a useful exercise: the empirical measure is now

$$\mu_t^N = \frac{1}{N} \sum_{i=1}^{N} \delta_{(x_i(t), v_i(t))},$$

and the weak formulation leads to a kinetic equation for $f(t, x, v)$.

1.2 Transport Preliminaries: Flows, Continuity Equations, and Duality

1.2.1 Overview

Before studying the nonlinear mean-field equation, we recall the basic transport and continuity equations. The goal is to understand how particle trajectories, flow maps, and measure-valued solutions fit together.

1. Linear transport equation. Consider $\partial_t u + b(t, x) \cdot \nabla_x u = 0$. Define a *characteristic curve* $X(t)$ as the solution of $\dot{X}(t) = b(t, X(t))$. Along this curve the total derivative of u vanishes:

$$\frac{d}{dt} u(t, X(t)) = \partial_t u + \dot{X} \cdot \nabla_x u = \partial_t u + b \cdot \nabla_x u = 0.$$

Hence the value of u is constant along the trajectory: to find $u(t, x)$, trace the flow backward to $t = 0$, $u(t, x) = u_0(\Phi(0, t, x))$.

2. Extension to mean-field characteristics. In the mean-field setting, the velocity field b is no longer prescribed but depends on the distribution ρ_t itself, e.g. $b[\rho_t] = K * \rho_t$. The method of characteristics still applies: the mass distribution evolves by the *push-forward* of the initial measure along the self-consistent trajectories, $\rho_t = \Phi_{t\#}\rho_0$.

1.2.2 Non-autonomous flow maps

Let $b : \mathbb{R} \times \mathbb{R}^d \to \mathbb{R}^d$ be a time-dependent velocity field. Consider the particle trajectory

$$\dot{z}(t) = b(t, z(t)), \qquad z(s) = x. \tag{3}$$

The associated flow map $\Phi(t, s, x)$ satisfies

$$\partial_t \Phi(t, s, x) = b(t, \Phi(t, s, x)), \qquad \Phi(s, s, x) = x. \tag{4}$$

Existence and uniqueness of the flow map.

Theorem 1.7 (Existence of the flow)**.** *Assume that $b \in C^1(\mathbb{R} \times \mathbb{R}^d; \mathbb{R}^d)$ and $|b(t, x)| \le C(1 + |x|)$ for all t, x. Then for every $s \in \mathbb{R}$ and $x \in \mathbb{R}^d$, there exists a global solution*

$$t \mapsto \Phi(t, s, x)$$

of (4).

Proof idea. Since $b \in C^1$, it is locally Lipschitz in the spatial variable. The Cauchy–Lipschitz theorem gives a unique local solution. The linear-growth assumption prevents finite-time blow-up: if $X(t) = \Phi(t, s, x)$, then

$$\frac{d}{dt}|X(t)| \le |b(t, X(t))| \le C(1 + |X(t)|).$$

By Gronwall, $|X(t)| \le (|x| + C|t - s|)e^{C|t-s|}$ on bounded intervals, so the local solution extends globally.

Theorem 1.8 (Uniqueness of the flow map)**.** *Assume $b(t, x)$ is globally Lipschitz in the spatial variable: there exists $L > 0$ such that*

$$|b(t, x) - b(t, y)| \le L|x - y| \qquad \text{for all } t \in \mathbb{R},\ x, y \in \mathbb{R}^d.$$

Then (4) *admits at most one solution.*

Proof. Suppose Φ_1, Φ_2 are two solutions with the same initial data (s, x). Using the integral form,

$$\Phi_1(t, s, x) - \Phi_2(t, s, x) = \int_s^t \big(b(r, \Phi_1) - b(r, \Phi_2)\big)\, dr,$$

so the Lipschitz condition gives

$$|\Phi_1(t, s, x) - \Phi_2(t, s, x)| \le L \int_s^t |\Phi_1(r, s, x) - \Phi_2(r, s, x)|\, dr.$$

Setting $\eta(t) = |\Phi_1(t, s, x) - \Phi_2(t, s, x)|$ and noting $\eta(s) = 0$, Gronwall's inequality gives $\eta(t) \le \eta(s)e^{L|t-s|} = 0$. Hence $\Phi_1 = \Phi_2$. $\square$

Proposition 1.9 (Algebraic properties of the flow). *The flow map satisfies:*

1. ***Composition (cocycle property):*** $\Phi(t_3, t_2, \Phi(t_2, t_1, x)) = \Phi(t_3, t_1, x)$.
2. ***Time reversibility:*** *the inverse of* $x \mapsto \Phi(t, s, x)$ *is* $x \mapsto \Phi(s, t, x)$.

Lemma 1.10 (Backward-flow identity). *Under the assumptions of the flow theorem, the map* $\Phi(s, t, x)$ *satisfies*

$$\partial_t \Phi(s, t, x) + D_x \Phi(s, t, x)\, b(t, x) = 0.$$

Proof idea. Use the identity $\Phi(s, t, \Phi(t, s, y)) = y$ and differentiate with respect to t.

Remark 1.11 (Determinism and flow). All these properties are consequences of the *deterministic property* of ODEs: the future is uniquely determined by the past, and a trajectory remains consistent regardless of where one restarts along the same path. This concept underlies Dobrushin's stability theorem: when comparing two trajectories, we trace the flow backward to control current discrepancies by the initial errors.

1.2.3 Method of Characteristics for the Transport Equation

Consider the non-autonomous transport equation

$$\partial_t u + b(t, x) \cdot \nabla_x u = 0, \qquad u(0, x) = g(x).$$

The operator $\partial_t + b(t, x) \cdot \nabla_x$ is the directional derivative in space-time along the vector field $(1, b(t, x))$. The characteristic curves are the integral curves of this space-time vector field; using t as the curve parameter gives $\dot{X}(t) = b(t, X(t))$.

Along such a curve,

$$\frac{d}{dt} u(t, X(t)) = \partial_t u(t, X(t)) + b(t, X(t)) \cdot \nabla_x u(t, X(t)) = 0.$$

Thus u is constant along characteristics. In terms of the flow,

$$u(t, x) = g(\Phi(0, t, x)).$$

Key message. The method of characteristics turns the transport PDE into an ODE along the characteristic curves.

1.2.4 Continuity Equation and Push-forward

The continuity equation associated with the same velocity field is

$$\partial_t \mu_t + \nabla \cdot (b(t, \cdot)\mu_t) = 0, \qquad \mu_{t=0} = \mu_0.$$

It describes the evolution of a mass distribution under the flow generated by b.

Definition 1.12 (Push-forward). Let $T : \mathbb{R}^d \to \mathbb{R}^d$ be measurable and $\mu \in \mathcal{P}(\mathbb{R}^d)$. The push-forward $T_\# \mu$ is defined by

$$(T_\# \mu)(A) = \mu(T^{-1}(A))$$

for every Borel set $A \subset \mathbb{R}^d$. Equivalently, for every bounded measurable test function φ,

$$\int_{\mathbb{R}^d} \varphi(y)\, (T_\# \mu)(dy) = \int_{\mathbb{R}^d} \varphi(T(x))\, \mu(dx).$$

Proposition 1.13 (Basic properties of push-forward). *Let $\mu \in \mathcal{P}(X)$, $T : X \to Y$, $S : Y \to Z$ be measurable. Then:*

1. *Composition: $(S \circ T)_\# \mu = S_\#(T_\# \mu)$.*
2. *Push-forward of an empirical measure:*
$$T_\# \left(\frac{1}{N} \sum_{i=1}^{N} \delta_{x_i} \right) = \frac{1}{N} \sum_{i=1}^{N} \delta_{T(x_i)}.$$
3. *Density transformation: if $\mu(dx) = f(x)\,dx$ and T is a C^1-diffeomorphism, then*
$$T_\# \mu(dy) = f(T^{-1}(y)) |\det DT^{-1}(y)|\, dy.$$

Theorem 1.14 (Push-forward solution of the continuity equation). *Let $\mu_0 \in \mathcal{P}(\mathbb{R}^d)$. The weak solution of*
$$\partial_t \mu_t + \nabla \cdot (b(t, \cdot)\mu_t) = 0, \qquad \mu_{t=0} = \mu_0,$$
is uniquely given by
$$\mu_t = \Phi(t, 0, \cdot)_\# \mu_0.$$
Equivalently, for every bounded measurable φ,
$$\int_{\mathbb{R}^d} \varphi(y)\, \mu_t(dy) = \int_{\mathbb{R}^d} \varphi(\Phi(t, 0, z))\, \mu_0(dz).$$

Proof. **Existence.** Let $\mu_t := \Phi(t, 0, \cdot)_\# \mu_0$. Then for every $\varphi \in C_c^1(\mathbb{R}^d)$,
$$\int_{\mathbb{R}^d} \varphi(y)\, \mu_t(dy) = \int_{\mathbb{R}^d} \varphi(\Phi(t, 0, z))\, \mu_0(dz).$$
Differentiating in time,
$$\frac{d}{dt} \int_{\mathbb{R}^d} \varphi(y)\, \mu_t(dy) = \int_{\mathbb{R}^d} \nabla\varphi(\Phi(t, 0, z)) \cdot b(t, \Phi(t, 0, z))\, \mu_0(dz) = \int_{\mathbb{R}^d} \nabla\varphi(y) \cdot b(t, y)\, \mu_t(dy),$$
which is the weak formulation of $\partial_t \mu_t + \nabla \cdot (b\mu_t) = 0$.

Uniqueness. Let μ_t be any distributional solution and define $\nu_t := \Phi(0, t, \cdot)_\# \mu_t$. For every $\varphi \in C_c^1(\mathbb{R}^d)$,
$$\langle \nu_t, \varphi \rangle = \int_{\mathbb{R}^d} \varphi(\Phi(0, t, z))\, \mu_t(dz).$$
Differentiating in time and using the backward-flow identity together with the weak formulation of the continuity equation, the two contributions cancel and $\frac{d}{dt}\langle \nu_t, \varphi \rangle = 0$. Hence $\nu_t = \nu_0 = \mu_0$, so $\mu_t = \Phi(t, 0, \cdot)_\# \mu_0$. □

1.2.5 Duality between Transport and Continuity

The transport and continuity equations describe the same flow from two dual viewpoints.

Let $\Phi(t, s, x)$ be the flow generated by $b(t, x)$. If u solves $\partial_t u + b \cdot \nabla u = 0$ with $u(0, x) = g(x)$, then $u(t, x) = g(\Phi(0, t, x))$. Equivalently, if $X(t) = \Phi(t, 0, a)$, then $u(t, X(t)) = g(a)$: the value of u is constant along forward characteristics, and to compute $u(t, x)$ at a fixed Eulerian point, one traces x backward to its initial footpoint $\Phi(0, t, x)$.

On the other hand, the continuity equation $\partial_t \mu_t + \nabla \cdot (b\mu_t) = 0$ moves mass forward along the flow: $\mu_t = \Phi(t, 0, \cdot)_\# \mu_0$.

In short: functions are pulled back by the inverse flow, while measures are pushed forward by the forward flow.

Adjoint formulation. Let $Au := b \cdot \nabla u$. With respect to the pairing $\langle u, \mu \rangle := \int u \, d\mu$, the formal adjoint is $A^*\mu = -\nabla \cdot (b\mu)$. Thus the transport equation reads $\partial_t u = -Au$, while the forward equation for measures reads $\partial_t \mu_t = A^* \mu_t$.

Conservation along the pairing. If u solves the transport equation and μ_t solves the continuity equation, then

$$\frac{d}{dt} \int_{\mathbb{R}^d} u(t,x) \, \mu_t(dx) = 0.$$

Indeed, writing $\mu_t(dx) = \rho(t,x) \, dx$ and using both equations,

$$\frac{d}{dt} \int u\rho \, dx = \int (\partial_t u)\rho \, dx + \int u \, \partial_t \rho \, dx = -\int (b \cdot \nabla u)\rho \, dx - \int u \, \nabla \cdot (b\rho) \, dx.$$

Integrating by parts in the second term gives $-\int u \, \nabla \cdot (b\rho) \, dx = \int (\nabla u \cdot b) \rho \, dx$, so the two contributions cancel.

Key message. The transport equation follows values along trajectories. The continuity equation transports mass along the same trajectories. These are two dual viewpoints on the same flow.

1.2.6 Empirical Measures as Weak Solutions

We use "weak solution" to mean *distributional solution*: tested against $\varphi \in C_c^\infty$. No Sobolev regularity is assumed, since μ_t^N is a sum of Dirac masses.

Apply the push-forward theorem to a finite particle system *without interaction*:

$$\dot{x}_i(t) = b(t, x_i(t)), \qquad x_i(0) = x_i^0, \qquad i = 1, \dots, N.$$

Define $\mu_t^N := \frac{1}{N} \sum_{i=1}^N \delta_{x_i(t)}$. Then

$$\mu_t^N = \Phi(t, 0, \cdot)_\# \mu_0^N, \qquad \mu_0^N = \frac{1}{N} \sum_{i=1}^N \delta_{x_i^0},$$

so μ_t^N solves $\partial_t \mu_t^N + \nabla \cdot (b\mu_t^N) = 0$ in the weak sense.

Proposition 1.15 (Empirical measure solves the continuity equation weakly)**.** *Let $x_1(t), \dots, x_N(t)$ solve $\dot{x}_i(t) = b(t, x_i(t))$ and define $\mu_t^N := \frac{1}{N} \sum_{i=1}^N \delta_{x_i(t)}$. Then μ_t^N satisfies $\partial_t \mu_t^N + \nabla \cdot (b(t, \cdot)\mu_t^N) = 0$ in the sense of distributions.*

Proof idea. For $\varphi \in C_c^1(\mathbb{R}^d)$,

$$\int \varphi \, d\mu_t^N = \frac{1}{N} \sum_i \varphi(x_i(t)), \quad \text{so} \quad \frac{d}{dt} \int \varphi \, d\mu_t^N = \frac{1}{N} \sum_i \nabla\varphi(x_i(t)) \cdot b(t, x_i(t)) = \int b \cdot \nabla\varphi \, d\mu_t^N,$$

which is the weak formulation of the continuity equation.

Remark 1.16. This computation is the first basic instance of the particle–PDE dictionary: particle trajectories solve ODEs, while their empirical measure solves a continuity equation.

Compactness viewpoint. Assume that for every $T > 0$, the particle trajectories remain in a common compact set: $\{x_i(t) : i = 1, \dots, N,\ t \in [0,T]\} \subset \Omega_T \subset \mathbb{R}^d$. Then $\operatorname{supp} \mu_t^N \subset \Omega_T$ for all $t \in [0,T]$. By Prokhorov's theorem, the family $(\mu_t^N)_N$ is tight, and along a subsequence $\mu_t^{N_k} \rightharpoonup \rho_t$. Passing to the limit in the weak formulation gives

$$\frac{d}{dt} \int \varphi \, d\rho_t = \int b \cdot \nabla \varphi \, d\rho_t,$$

i.e. $\partial_t \rho + \nabla \cdot (b\rho) = 0$ in the weak sense.

> **Key message.** Compactness gives subsequential limits. To identify the full limit, one needs uniqueness or stability.

1.3 Mean-Field Characteristics

1.3.1 Interaction Kernel and Particle System

We now consider an interaction kernel

$$K : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}^d.$$

The assumptions used in the smooth Dobrushin theory are global Lipschitz regularity and at most linear growth. In many physical examples one also has an antisymmetry condition.

(A1) **Optional antisymmetry:** $K(z, z') = -K(z', z)$. This expresses Newton's third-law structure (action–reaction) and implies $K(z, z) = 0$. It is useful for conservation laws and for many physical models, but it is not needed for the Dobrushin stability estimate.

(A2) **Regularity:** $K \in C^1(\mathbb{R}^d \times \mathbb{R}^d; \mathbb{R}^d)$.

(A3) **Bounded first derivatives:** there exists $L > 0$ such that

$$\sup_{z,z'} |\nabla_z K(z, z')| \le L, \qquad \sup_{z,z'} |\nabla_{z'} K(z, z')| \le L.$$

Assumption (A3) implies that K is globally Lipschitz in both variables:

$$|K(z_1, z') - K(z_2, z')| \le L|z_1 - z_2|, \qquad |K(z, z_1') - K(z, z_2')| \le L|z_1' - z_2'|.$$

It also gives at most linear growth: $|K(z, z')| \le C(1 + |z| + |z'|)$.

For $\mu \in \mathcal{P}_1(\mathbb{R}^d)$, define

$$(K * \mu)(z) := \int_{\mathbb{R}^d} K(z, z') \, \mu(dz').$$

Under the assumptions above, $K * \mu$ is well-defined and globally Lipschitz in z.

The N-particle system is

$$\dot{z}_i(t) = \frac{1}{N} \sum_{j=1}^{N} K(z_i(t), z_j(t)), \qquad z_i(0) = z_i^0, \qquad i = 1, \dots, N.$$

Theorem 1.17 (Well-posedness of the particle system)**.** *Assume K satisfies (A2)–(A3). Then, for every initial condition $(z_1^0, \dots, z_N^0) \in (\mathbb{R}^d)^N$, the particle system admits a unique global solution $(z_1(t), \dots, z_N(t)) \in C^1([0, \infty); (\mathbb{R}^d)^N)$.*

Proof idea. The vector field

$$B_N(z_1,\dots,z_N) = \left(\frac{1}{N}\sum_{j=1}^{N} K(z_1,z_j),\dots,\frac{1}{N}\sum_{j=1}^{N} K(z_N,z_j)\right)$$

is globally Lipschitz on $(\mathbb{R}^d)^N$. The claim follows from the Cauchy–Lipschitz theorem.

1.3.2 Mean-Field PDE and Mean-Field Characteristics

Assume now that $K : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}^d$ is globally Lipschitz with at most linear growth. The expected limiting equation is the nonlinear continuity equation

$$\partial_t \rho_t + \nabla \cdot \big(\rho_t (K * \rho_t)\big) = 0, \qquad \rho_{|t=0} = \rho_0. \tag{5}$$

In weak form, for every $\varphi \in C_c^1(\mathbb{R}^d)$,

$$\frac{d}{dt}\int_{\mathbb{R}^d} \varphi(z)\,\rho_t(dz) = \int_{\mathbb{R}^d} \nabla\varphi(z)\cdot (K*\rho_t)(z)\,\rho_t(dz). \tag{6}$$

The motivation for the mean-field characteristic flow mirrors the linear case. If the curve $(\rho_t)_t$ were already known, then $b(t,z) := (K*\rho_t)(z)$ would be an ordinary time-dependent velocity field, and the characteristics would solve

$$\dot Z(t;a) = (K*\rho_t)(Z(t;a)), \qquad Z(0;a) = a.$$

The new feature is the self-consistency condition

$$\rho_t = Z(t,\cdot)_{\#}\rho_0.$$

Equivalently, for every test function φ,

$$\int_{\mathbb{R}^d} \varphi(z)\,\rho_t(dz) = \int_{\mathbb{R}^d} \varphi(Z(t;a))\,\rho_0(da).$$

Using the push-forward relation, the characteristic equation can also be written as

$$\dot Z(t;a) = \int_{\mathbb{R}^d} K\big(Z(t;a), Z(t;a')\big)\,\rho_0(da'). \tag{7}$$

This highlights the mean-field structure: each trajectory evolves under the average interaction generated by the full distribution of particles.

Remark 1.18. This is an ODE for a single typical particle, not for an N-particle configuration. The nonlinearity enters through the law ρ_t, which must be determined self-consistently from the same characteristic flow.

Theorem 1.19 (Well-posedness and characteristic representation)**.** *Assume that K is globally Lipschitz in both variables with constant L and has at most linear growth. Then, for every $\rho_0 \in \mathcal{P}_1(\mathbb{R}^d)$ and every $T > 0$, there exists a unique curve*

$$\rho \in C([0,T];\mathcal{P}_1(\mathbb{R}^d))$$

solving (5) *in the weak sense* (6)*. Moreover, there exists a unique characteristic flow $Z(t;a)$ satisfying $\dot Z(t;a) = (K*\rho_t)(Z(t;a))$ with $Z(0;a) = a$, and the solution is represented by $\rho_t = Z(t,\cdot)_{\#}\rho_0$. Finally, the solution map is stable in W_1: if ρ_t and $\tilde\rho_t$ are two solutions with initial data $\rho_0, \tilde\rho_0 \in \mathcal{P}_1(\mathbb{R}^d)$, then*

$$W_1(\rho_t,\tilde\rho_t) \le e^{2Lt} W_1(\rho_0,\tilde\rho_0), \qquad 0 \le t \le T. \tag{8}$$

Proof idea. For a prescribed curve $(\mu_t)_{t\in[0,T]}$, the vector field $z \mapsto (K * \mu_t)(z)$ is globally Lipschitz in z, uniformly in time, with at most linear growth. The Cauchy–Lipschitz theorem gives a unique flow $Z^\mu(t;\cdot)$. This defines a map on curves of measures by

$$\mu_\cdot \mapsto Z^\mu(t,\cdot)_\# \rho_0,$$

which is a contraction on a sufficiently short time interval; the argument iterates to cover $[0,T]$.

The stability estimate follows by coupling two initial data optimally and transporting the coupling along the two characteristic flows. For paired characteristics $Z(t)$ and $\tilde{Z}(t)$, the Lipschitz assumption gives

$$\frac{d}{dt}|Z(t) - \tilde{Z}(t)| \le L|Z(t) - \tilde{Z}(t)| + LW_1(\rho_t, \tilde{\rho}_t).$$

Averaging over the initial coupling and Gronwall give (8).

Key message. Mean-field characteristics are ordinary characteristics for a continuity equation whose velocity field is determined self-consistently by the evolving probability measure.

Remark 1.20 (Beyond globally Lipschitz kernels)*.* The global Lipschitz assumption gives a clean Cauchy–Lipschitz theory and a direct Dobrushin stability estimate. It is not the endpoint of the theory. Mean-field limits have been proved under rougher assumptions, including one-sided Lipschitz or monotonicity conditions, Sobolev or L^p-type regularity assumptions, and several classes of singular interaction kernels. In those settings the characteristic flow may need to be replaced by a regular Lagrangian flow, an entropy formulation, a modulated energy estimate, or a compactness argument.

1.3.3 Empirical Initial Data Recover the Particle System

The well-posedness theorem applies to arbitrary probability measures in $\mathcal{P}_1(\mathbb{R}^d)$, including atomic measures. This is essential because the empirical measure of an N-particle configuration is an atomic probability measure

$$\mu_0^N = \frac{1}{N}\sum_{i=1}^N \delta_{z_i^0}.$$

The deterministic N-particle system can therefore be viewed as a special mean-field characteristic solution whose initial law is empirical. This observation is the bridge between the characteristic formulation and Dobrushin's stability estimate: once empirical initial data are allowed, the same mean-field solution map can evolve both the continuum initial law ρ_0 and the discrete empirical law μ_0^N.

Proposition 1.21 (Empirical initial data recover the particle system)**.** *Let $z_1^0, \dots, z_N^0 \in \mathbb{R}^d$ and set $\mu_0^N := \frac{1}{N}\sum_{i=1}^N \delta_{z_i^0}$. Let μ_t^N be the unique mean-field solution of*

$$\partial_t \mu_t^N + \nabla \cdot \left(\mu_t^N (K * \mu_t^N)\right) = 0, \qquad \mu^N_{|t=0} = \mu_0^N.$$

Then μ_t^N remains empirical:

$$\mu_t^N = \frac{1}{N}\sum_{i=1}^N \delta_{z_i(t)},$$

where $z_1(t), \dots, z_N(t)$ solve the particle system

$$\dot z_i(t) = \frac{1}{N}\sum_{j=1}^N K(z_i(t), z_j(t)), \qquad z_i(0) = z_i^0.$$

Conversely, if $z_1(t), \dots, z_N(t)$ solve this particle system, then $\mu_t^N := \frac{1}{N}\sum_{i=1}^N \delta_{z_i(t)}$ solves the mean-field continuity equation in the weak sense.

Proof. Let $Z(t; \cdot, \mu_0^N)$ be the characteristic flow associated with the initial law μ_0^N, and define $z_i(t) := Z(t; z_i^0, \mu_0^N)$. By the characteristic representation,

$$\mu_t^N = Z(t; \cdot, \mu_0^N)_\# \mu_0^N = \frac{1}{N}\sum_{j=1}^N \delta_{z_j(t)}.$$

Therefore

$$\dot z_i(t) = (K * \mu_t^N)(z_i(t)) = \int_{\mathbb{R}^d} K(z_i(t), z')\,\mu_t^N(dz') = \frac{1}{N}\sum_{j=1}^N K(z_i(t), z_j(t)).$$

This gives the particle system.

Conversely, if $z_i(t)$ solve the particle system and $\mu_t^N = N^{-1}\sum_i \delta_{z_i(t)}$, then for every $\varphi \in C_c^1(\mathbb{R}^d)$,

$$\frac{d}{dt}\int_{\mathbb{R}^d} \varphi(z)\,\mu_t^N(dz) = \frac{1}{N}\sum_{i=1}^N \nabla\varphi(z_i(t))\cdot \dot z_i(t) = \int_{\mathbb{R}^d} \nabla\varphi(z)\cdot (K * \mu_t^N)(z)\,\mu_t^N(dz),$$

which is the weak form of the mean-field continuity equation. □

Key message. The empirical particle system is the mean-field characteristic flow started from empirical initial data. What remains is to show stability with respect to the initial measure.

1.3.4 Why Wasserstein Distance, and Why W_1?

To compare μ_t^N and ρ_t, the natural metric is not an L^p-distance between densities but a transportation distance between measures.

Roughly speaking, L^p-type quantities compare mass *vertically* at the same location: $\|f - g\|_{L^p}$. Wasserstein distances compare mass *horizontally*, by measuring how far mass must be transported from one location to another.

This is essential for empirical measures. The empirical measure μ_t^N is singular with respect to Lebesgue measure, while the limiting law ρ_t may be smooth. L^p-comparison is therefore not appropriate. Total variation is also usually too strong: if ρ_t is non-atomic, then μ_t^N and ρ_t are mutually singular for every finite N, so their total variation distance does not go to zero. Transport distances are designed precisely for this situation.

Definition 1.22 (Wasserstein-1 distance)**.** For $\mu, \nu \in \mathcal{P}_1(\mathbb{R}^d)$,

$$W_1(\mu, \nu) := \inf_{\pi \in \Pi(\mu,\nu)} \iint_{\mathbb{R}^d\times\mathbb{R}^d} |x - y|\,\pi(dx, dy),$$

where $\Pi(\mu, \nu)$ is the set of all couplings of μ and ν.

There are two basic reasons W_1 is the classical distance in Dobrushin's argument.

1. Lipschitz vector fields are naturally controlled by W_1. If K is Lipschitz in its second variable, then for fixed x, $y \mapsto K(x,y)$ is a Lipschitz function. For any coupling $\pi \in \Pi(\rho, \tilde{\rho})$,

$$\begin{aligned}|(K*\rho)(x) - (K*\tilde{\rho})(x)| &= \left| \iint \big(K(x,y) - K(x,\tilde{y})\big)\, \pi(dy, d\tilde{y}) \right| \\ &\le L \iint |y - \tilde{y}|\, \pi(dy, d\tilde{y}).\end{aligned}$$

Taking the infimum over couplings, $|(K*\rho)(x) - (K*\tilde{\rho})(x)| \le L W_1(\rho, \tilde{\rho})$. Thus W_1 directly controls the difference between the mean-field velocity fields.

2. W_1 is dual to Lipschitz observables. The Kantorovich–Rubinstein duality says

$$W_1(\mu, \nu) = \sup_{\mathrm{Lip}(\varphi) \le 1} \left| \int_{\mathbb{R}^d} \varphi \, d\mu - \int_{\mathbb{R}^d} \varphi \, d\nu \right|.$$

Convergence in W_1 means that empirical averages converge uniformly against all 1-Lipschitz observables: $\frac{1}{N}\sum_{i=1}^N \varphi(z_i) \approx \int_{\mathbb{R}^d} \varphi(z)\, \rho(dz)$. This matches the weak formulation of the continuity equation and the Lipschitz structure of the interaction kernel. In this sense, W_1 is the transport distance naturally paired with first-order, Lipschitz information.

Remark 1.23 (Why not state everything in W_2?). For globally Lipschitz kernels a Dobrushin-type argument can also be written in W_p, including W_2, provided the corresponding p-th moments are finite. For example, one may estimate $\iint |Z(t;a) - \tilde{Z}(t;b)|^p \, \gamma_0(da, db)$ and close the estimate by the same Lipschitz bound and Gronwall's lemma.

The reason W_1 is the classical and minimal formulation is that it only requires finite first moments and has a simple dual characterization by Lipschitz test functions. Higher Wasserstein distances require higher moment assumptions and do not have the same elementary dual formulation. Thus W_1 is not the only possible distance in the smooth theory, but it is the lowest-moment transport distance for which the standard Lipschitz stability argument closes cleanly.

A separate question is why Sznitman's synchronous coupling for stochastic particle systems naturally lives in W_2 rather than W_1. That is a genuinely different phenomenon, driven by Itô calculus and i.i.d. fluctuations rather than by the Lipschitz structure alone; we discuss it in Section 1.4.5.

Remark 1.24. A coupling $\pi \in \Pi(\mu, \nu)$ tells us how to match mass distributed according to μ with mass distributed according to ν. Dobrushin's argument works by transporting an initial coupling along two characteristic flows.

1.4 Dobrushin's Theorem and Applications

1.4.1 Dobrushin's Stability Estimate

Theorem 1.25 (Dobrushin stability in W_1)**.** *Assume that $K : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}^d$ is globally Lipschitz in both variables with Lipschitz constant L:*

$$|K(x,y) - K(x',y')| \le L|x - x'| + L|y - y'|.$$

Let $(\rho_t)_{t \ge 0}$ and $(\tilde{\rho}_t)_{t \ge 0}$ be two weak solutions of

$$\partial_t \rho_t + \nabla \cdot (\rho_t b[\rho_t]) = 0, \qquad b[\rho](x) = \int_{\mathbb{R}^d} K(x,y)\, \rho(dy),$$

with finite first moments. Then

$$W_1(\rho_t, \tilde{\rho}_t) \le e^{2Lt} W_1(\rho_0, \tilde{\rho}_0).$$

Proof. Let $\gamma_0 \in \Pi(\rho_0, \tilde{\rho}_0)$ be a coupling of the initial data. Define

$$T_t(a,b) := \big(Z(t;a,\rho_0), \tilde{Z}(t;b,\tilde{\rho}_0)\big), \qquad \gamma_t := (T_t)_{\#}\gamma_0,$$

so that $\gamma_t \in \Pi(\rho_t, \tilde{\rho}_t)$. Define the transportation cost along this coupling:

$$E(t) := \iint_{\mathbb{R}^d \times \mathbb{R}^d} |x-y| \, \gamma_t(dx, dy) = \iint |Z(t;a,\rho_0) - \tilde{Z}(t;b,\tilde{\rho}_0)| \, \gamma_0(da, db).$$

Using the characteristic equation,

$$Z(t;a,\rho_0) = a + \int_0^t \int_{\mathbb{R}^d} K(Z(s;a,\rho_0), Z(s;a',\rho_0)) \, \rho_0(da') \, ds.$$

Since γ_0 has first marginal ρ_0, this can be written as

$$Z(t;a,\rho_0) = a + \int_0^t \iint K(Z(s;a,\rho_0), Z(s;a',\rho_0)) \, \gamma_0(da', db') \, ds,$$

and similarly for $\tilde{Z}(t;b,\tilde{\rho}_0)$ with marginal $\tilde{\rho}_0$. Subtracting and using the Lipschitz property of K,

$$\begin{aligned}
&|Z(t;a,\rho_0) - \tilde{Z}(t;b,\tilde{\rho}_0)| \\
&\le |a-b| + L \int_0^t |Z(s;a,\rho_0) - \tilde{Z}(s;b,\tilde{\rho}_0)| \, ds \\
&\quad + L \int_0^t \iint |Z(s;a',\rho_0) - \tilde{Z}(s;b',\tilde{\rho}_0)| \, \gamma_0(da', db') \, ds.
\end{aligned}$$

Integrating with respect to $\gamma_0(da, db)$,

$$E(t) \le E(0) + 2L \int_0^t E(s) \, ds.$$

Gronwall gives $E(t) \le e^{2Lt} E(0)$. Since γ_t is a coupling of ρ_t and $\tilde{\rho}_t$, we have $W_1(\rho_t, \tilde{\rho}_t) \le E(t)$. Taking the infimum over initial couplings yields

$$W_1(\rho_t, \tilde{\rho}_t) \le e^{2Lt} W_1(\rho_0, \tilde{\rho}_0).$$

□

DK:[The N-particle ODE and the mean-field PDE are not two separate things: they are the same nonlinear continuity equation with different initial conditions. Convergence of particles to the PDE is then a stability statement with respect to the initial measure.]

1.4.2 A Standard Mean-Field Limit Theorem

Theorem 1.26 (Mean-field limit under globally Lipschitz interactions)**.** *Assume $K : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}^d$ is globally Lipschitz in both variables. Let $(\rho_t)_{t \ge 0}$ be the weak solution of*

$$\partial_t \rho_t + \nabla \cdot (\rho_t (K * \rho_t)) = 0$$

with initial datum $\rho_0 \in \mathcal{P}_1(\mathbb{R}^d)$*. Let* $X_1^N, \dots, X_N^N$ *solve*

$$\dot{X}_i^N(t) = \frac{1}{N}\sum_{j=1}^N K(X_i^N(t), X_j^N(t)), \qquad i = 1, \dots, N,$$

and define $\mu_t^N := \frac{1}{N}\sum_{i=1}^N \delta_{X_i^N(t)}$*. Then for every* $t \geq 0$*,*

$$W_1(\mu_t^N, \rho_t) \leq e^{2Lt} W_1(\mu_0^N, \rho_0).$$

In particular, if $W_1(\mu_0^N, \rho_0) \to 0$*, then* $W_1(\mu_t^N, \rho_t) \to 0$ *uniformly on every bounded time interval.*

Proof. Apply Dobrushin's stability estimate with $\tilde{\rho}_t = \mu_t^N$. By Proposition 1.21, μ_t^N is itself a weak solution of the same continuity equation, with empirical initial datum μ_0^N. Therefore

$$W_1(\mu_t^N, \rho_t) \leq e^{2Lt} W_1(\mu_0^N, \rho_0).$$ □

Remark 1.27. If the initial positions $X_i^{(0)}$ are i.i.d. with law ρ_0, then $\mu_0^N \to \rho_0$ almost surely in a suitable weak sense. Under standard moment assumptions, one also has convergence in W_1. Hence Dobrushin's estimate gives the mean-field limit directly from the approximation of the initial empirical measure.

Key message. Dobrushin turns the mean-field limit into a stability statement for the continuum equation: mean-field convergence follows from continuous dependence of the solution on the initial data in W_1.

1.4.3 Application: The Aggregation Equation

We now apply the previous theory to the aggregation equation

$$\partial_t \rho = \nabla \cdot \big(\rho \, \nabla W * \rho\big), \qquad \rho_{|t=0} = \rho^0,$$

where $W : \mathbb{R}^d \to \mathbb{R}$ is an interaction potential and $(\nabla W * \rho)(x) = \int_{\mathbb{R}^d} \nabla W(x-y)\,\rho(dy)$.

The corresponding particle system is

$$\dot{x}_i(t) = -\frac{1}{N}\sum_{j=1}^N \nabla W(x_i(t) - x_j(t)), \qquad x_i(0) = x_i^0.$$

This is the previous mean-field framework with $K(x,y) = -\nabla W(x-y)$.

Example 1 (Animal aggregation)*.* Here $x_i(t)$ is the position of individual i. The interaction potential W encodes attraction and repulsion: short-range repulsion prevents overcrowding, while long-range attraction keeps the group together.

Example 2 (Opinion dynamics)*.* Here $x_i(t)$ is the opinion of individual i in an opinion space. Interactions represent how individuals adjust their opinions after exposure to others.

Assumptions on W. Assume $W(x) = W(-x)$ and $W \in C_b^2(\mathbb{R}^d)$, so that ∇W is globally Lipschitz. Then $K(x,y) = -\nabla W(x-y)$ satisfies the smoothness and Lipschitz assumptions used above.

For the empirical measure $\mu_t^N = \frac{1}{N}\sum_{i=1}^N \delta_{x_i(t)}$, let

$$v(t,x) := -(\nabla W * \mu_t^N)(x) = -\frac{1}{N}\sum_{i=1}^N \nabla W(x - x_i(t)).$$

Then μ_t^N solves $\partial_t \mu_t^N + \nabla \cdot (v(t,\cdot)\mu_t^N) = 0$, or equivalently $\partial_t \mu_t^N = \nabla \cdot (\mu_t^N \nabla W * \mu_t^N)$.

Characteristics for the aggregation equation. The mean-field characteristics are

$$\dot{Z}(t; y, \rho^0) = -(\nabla W * \rho_t)(Z(t; y, \rho^0)), \qquad \rho_t = Z(t; \cdot, \rho^0)_\# \rho^0.$$

Theorem 1.28 (Solution by characteristics)**.** *Under the smoothness assumptions on W, the unique solution of the aggregation equation is $\rho_t = Z(t; \cdot, \rho^0)_\# \rho^0$.*

Stability for the aggregation equation. Let $\rho_1^0, \rho_2^0 \in \mathcal{P}_1(\mathbb{R}^d)$, and let $\rho_1(t), \rho_2(t)$ be the corresponding solutions. Then

$$W_1(\rho_1(t), \rho_2(t)) \leq e^{Ct} W_1(\rho_1^0, \rho_2^0),$$

with C depending on $\|D^2 W\|_{L^\infty}$.

Proof idea. Direct application of Dobrushin's estimate with $K(x, y) = -\nabla W(x - y)$. Since $W \in C_b^2(\mathbb{R}^d)$, K is globally Lipschitz with constant controlled by $\|D^2 W\|_{L^\infty}$; the general stability estimate gives the conclusion.

Corollary 1 (Mean-field limit for aggregation). Let μ_0^N be empirical measures satisfying $W_1(\mu_0^N, \rho^0) \to 0$. Let μ_t^N be generated by the discrete aggregation system. Then for each fixed $t \geq 0$, $W_1(\mu_t^N, \rho_t) \to 0$, where ρ_t solves $\partial_t \rho_t = \nabla \cdot (\rho_t \nabla W * \rho_t)$.

1.4.4 Remark: Singular Potentials

The smooth theory above relies on ∇W being globally Lipschitz. If $W(x) \sim |x|^\alpha$ near the origin, then $D^2 W$ is singular for $\alpha < 2$, and even ∇W is singular for $\alpha < 1$. Two difficulties then appear:

1. the particle system may fail to be globally well-posed because particles can collide;
2. the continuum equation may not be well-defined for measure-valued data because the convolution $\nabla W * \mu$ may not make sense.

One common strategy is to work with more regular initial data, for example $\rho^0 \in L^1 \cap L^p$ with p sufficiently large depending on the singularity of W, and to choose the particle approximation carefully so that collisions are avoided on the time interval of interest.

Remark 1.29. For mildly singular or λ-convex potentials, one can still obtain well-posedness and stability in Wasserstein distance. For more singular potentials, such as Coulomb or vortex-type kernels, Dobrushin's Lipschitz argument no longer applies directly and more refined methods are needed.

1.4.5 Remark: Synchronous Coupling and Propagation of Chaos

A common probabilistic route is to couple the particle system and the nonlinear process using the same randomness. In the deterministic setting, this means matching initial data. In the stochastic setting, it means using the same Brownian motions:

$$d\bar{X}_i(t) = b[\rho_t](\bar{X}_i(t))\, dt + \sqrt{2\sigma}\, dW_i(t).$$

One then writes an equation for $X_i^N(t) - \bar{X}_i(t)$. Under Lipschitz assumptions, the estimate closes by Gronwall.

Dobrushin is presented as a first-moment stability method at the level of measures, hence the natural W_1 formulation. Sznitman's synchronous coupling, while structurally analogous, naturally lives in L^2 and W_2 for two complementary reasons:

(1) Brownian cancellation. Driving both X_i^N and $\bar{X}_i$ by the same Brownian motion makes the stochastic integrals cancel in the difference $d(X_i^N - \bar{X}_i)$, reducing the SDE to a random ODE. This is the analog of "matching initial data" in the deterministic setting.

(2) Fluctuation control via second moments. Even after Brownian cancellation, the drift difference contains a fluctuation term $b[\bar{\mu}_t^N](\bar{X}_i) - b[\rho_t](\bar{X}_i)$, where $\bar{\mu}_t^N$ is the empirical measure of i.i.d. samples. This is exactly an i.i.d. centered sum, and L^2 is the natural setting for its CLT-type bound $|\cdot|^2 = O(1/N)$. Together with the smoothness of $|x|^2$ (vs. the non-differentiability of $|x|$ at zero), this is what makes Sznitman's argument cleanly close in W_2.

In settings where Brownian motion does *not* cancel pathwise (e.g. relative-entropy or Girsanov methods), Itô's formula plays a different role: it produces martingale terms that vanish only after taking expectation. That is a different mechanism, used in Jabin–Wang-style entropy methods.